Network Resource Management Based on Language Model Multi-Agent Systems

Zheyun Zhao

Mid-Term Review

BUPT No: 2022213670 QMUL No: 221170559

March 13, 2026

Outline

Background & Motivation

Methodology

Experimental Results

Problems, Solutions & Future Work

Research Background · Existing Challenges · Motivation

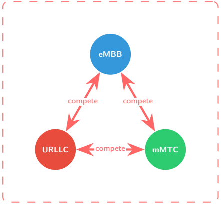
5G Network Slicing creates multiple logical networks over shared infrastructure. Three slice types share limited spectrum with **competing** QoS (Quality of Service) demands: eMBB (high throughput) / URLLC (ultra-low latency) / mMTC (massive access). Traditional rule-based and DRL (Deep Reinforcement Learning) methods struggle to unify 3GPP SLA (Service Level Agreement) constraints, operator policies, and dynamic user behaviour into an adaptive framework.

	Description
Gap 1	End-to-end multi-agent LLM (Large Language Model) framework for slicing remains underexplored
Gap 2	Competitive negotiation for constrained resource allocation is unaddressed by cooperative MAS (Multi-Agent System)
RQ1	Can multi-agent LLM outperform rule-based and single-agent baselines?
RQ2	Does negotiation provide structural advantages over direct allocation?

eMBB (Enhanced Mobile Broadband): high-throughput services, e.g. video streaming.
URLLC (Ultra-Reliable Low-Latency Comm.): mission-critical apps, sub-ms latency.

mMTC (Massive Machine-Type Comm.): massive IoT device connectivity.

Framework	Architecture	Multi-Agent	Negotiation	E2E Slicing
WirelessAgent [**tong2025wirelessagent**]	Percept-Memory-Plan-Act	✗	✗	✗
CommLLM [**jiang2024commllm**]	Three-component MAS	✓	✗	✗ Semantic
Qu et al. [**qu2025dualloop**]	Dual-loop edge-terminal	✓	Centralised	Partial
MetaGPT [**hong2023metagpt**] / ChatDev [**qian2023chatdev**]	Role-based cooperative	✓	✓ Cooperative	✗ General
This Work	**Competitive MAS**	✓	✓ **Competitive**	✓

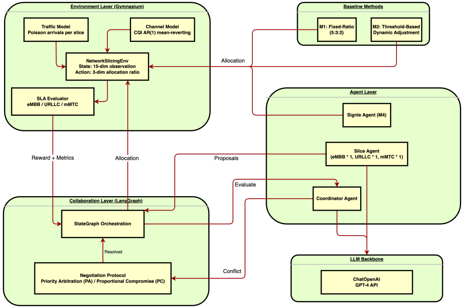


Fixed B_{total}, zero-sum inter-slice competition

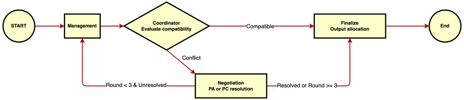
Attemps: (1) Reproducible Gymnasium-based 5G slicing simulation; (2) LangGraph-orchestrated two-tier MAS framework with proposal-evaluation-negotiation protocol.

Methodology: Architecture · Two-Tier Scheduling · Negotiation

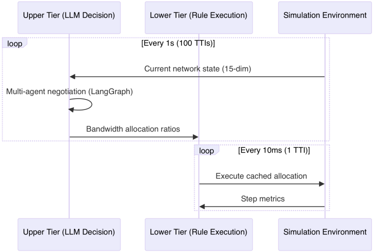
System Architecture — Agent Layer / Collaboration Layer (LangGraph) / Environment Layer (Gymnasium):



Negotiation Flow (LangGraph StateGraph, max 3 rounds):



Two-Tier Scheduling — Bridging LLM inference latency (seconds) vs network real-time requirements (milliseconds):

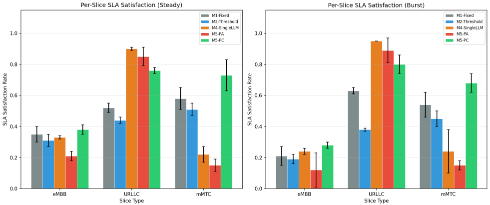


ID	Method	Category	Description
M1	Fixed-Ratio	Rule-based	Fixed 5:3:2 allocation
M2	Threshold	Rule-based	SLA-triggered dynamic adjustment
M4	Single-Agent LLM	Single LLM	GPT-4 + CoT (Chain-of-Thought)
M5-PA	Multi-Agent PA	Multi LLM	Priority arbitration negotiation
M5-PC	Multi-Agent PC	Multi LLM	Proportional compromise negotiation

Experiment 1: Core Performance Comparison (RQ1)

Steady-State Scenario (100 users, 100 decision cycles × 2 seeds):

Method	SLA$_{\text{eMBB}}$	SLA$_{\text{URLLC}}$	SLA$_{\text{mMTC}}$	$\bar{S}$	Fairness	Lat. (ms)	Tokens
M1-Fixed	0.35±0.05	0.52±0.03	0.58±0.07	0.48	0.893	0	0
M2-Threshold	0.31±0.04	0.44±0.02	0.51±0.04	0.42	0.912	0	0
M4-SingleLLM	0.33±0.01	**0.90**±0.01	0.22±0.05	0.48	0.720	6326	26k
M5-PA	0.21±0.03	0.85±0.06	0.15±0.04	0.40	0.616	8065	125k
M5-PC	0.38±0.03	0.76±0.02	**0.73**±0.10	**0.62**	**0.952**	8033	125k



Burst Scenario (eMBB users doubled at cycle 50):

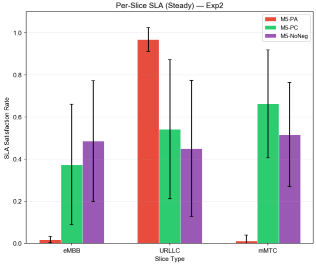
Method	SLA$_{\text{eMBB}}$	SLA$_{\text{URLLC}}$	SLA$_{\text{mMTC}}$	$\bar{S}$	Fairness	Lat. (ms)	Tokens
M1-Fixed	0.21±0.06	0.63±0.02	0.54±0.08	0.46	0.831	0	0
M2-Threshold	0.19±0.03	0.38±0.01	0.45±0.05	0.34	0.876	0	0
M4-SingleLLM	0.24±0.02	**0.95**±0.00	0.24±0.14	0.48	0.660	6063	26k
M5-PA	0.12±0.11	0.89±0.08	0.15±0.03	0.38	0.541	7957	125k
M5-PC	0.28±0.02	0.80±0.06	**0.68**±0.06	**0.59**	**0.924**	7887	125k

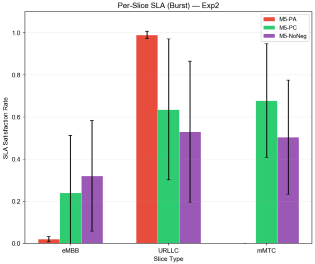
Key Findings (RQ1):

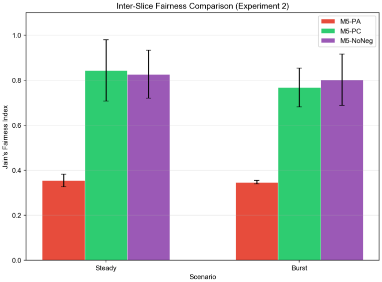
- **M5-PC** outperforms all baselines in overall SLA ($\bar{S}$=0.62) with Fairness=0.952, far above M4 (0.720) and M5-PA (0.616)
- M5-PA over-prioritises URLLC at the expense of eMBB/mMTC
- M4 tends toward extreme allocation: very high URLLC (0.90) but very low mMTC (0.22)
- Rule methods have zero latency but cannot reason about SLA semantics
- **Proportional compromise is the best strategy for balancing SLA and fairness**

Experiment 2: Negotiation Mechanism Validation (RQ2)

Method	Scenario	SLA_{eMBB}	SLA_{URLLC}	SLA_{mMTC}	$\bar{S}$	Fairness
M5-PA	steady	0.21	0.85	0.15	0.40	0.616
M5-PA	burst	0.12	0.89	0.15	0.38	0.541
M5-PC	steady	0.38	0.76	0.73	**0.62**	**0.952**
M5-PC	burst	0.28	0.80	0.68	**0.59**	**0.924**
M5-NoNeg	steady	0.56	0.49	0.14	0.39	0.810
M5-NoNeg	burst	0.32	0.71	0.36	0.46	0.839







Key Findings (RQ2) — Structural Advantage of Negotiation:

- **M5-PC vs M5-NoNeg (steady)**: $\bar{S}$ +59% (0.39→0.62), Fairness +18% (0.81→0.95)
- **M5-PC vs M5-PA**: Proportional compromise outperforms priority arbitration across all metrics, especially mMTC SLA (0.73 vs 0.15)
- M5-NoNeg uses LLM but lacks iterative negotiation, resulting in imbalanced allocation (eMBB=0.56, mMTC=0.14)
- M5-PA hard priority causes URLLC to monopolise resources (0.85+); low-priority slices are systematically under-served
- **Conclusion**: The negotiation protocol itself (not merely more LLM calls) drives significant SLA and fairness improvements

Problems & Solutions · Future Work

Identified Problems

- **MAS latency incompatible with real-time control.** Multi-agent negotiation takes $\sim$8 s per step, far exceeding the millisecond-level response time required by network control loops.
- **Statistical significance.** Current experiments use only 2 random seeds with 100 decision cycles per episode, which limits the robustness of conclusions.

Proposed Solutions

- **Strategic MAS + tactical rule executor** *(in progress)*: Reposition MAS as a **strategic layer** that predicts traffic trends and generates allocation *policies*, rather than making per-TTI decisions. The lower-tier rule executor operates at 10 ms granularity guided by MAS-provided policies, decoupling LLM inference from the real-time control loop.
- **Expanded experimental scale**: Increase to 5 random seeds $\times$ 500 decision cycles per episode to strengthen statistical validity.

Future Work

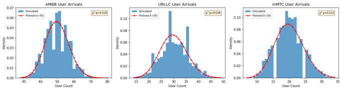
- **Strategic MAS + tactical rule executor**: Validate the two-tier paradigm where MAS provides medium-term allocation policies and the rule layer handles real-time execution
- **DRL baseline (M3-PPO)**: Train PPO (Proximal Policy Optimisation) with Stable-Baselines3 as an additional baseline
- **RAG (Retrieval-Augmented Generation) integration**: Inject 3GPP specifications and telecom domain knowledge into agent prompts
- **Agent memory**: Leverage historical allocation patterns to improve decision quality over time

Setup: 100 MHz (NR n78), 100 users, GPT-4o-mini (temp=0), negotiation $\leq$ 3 rounds, Gymnasium + LangGraph.

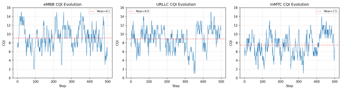
References

Thank You!

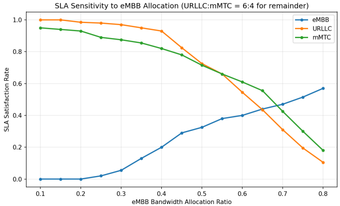
Appendix A: Simulation Environment Validation



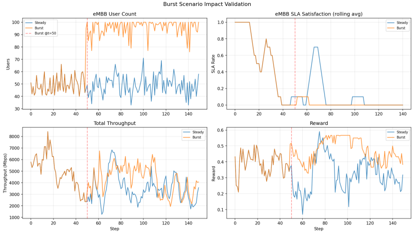
Poisson arrival validation (χ^2 test)



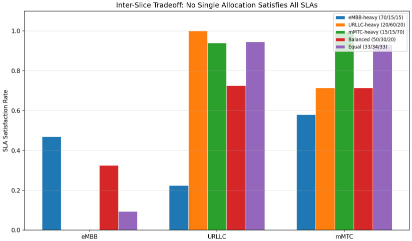
CQI AR(1) mean-reverting evolution



SLA sensitivity to allocation

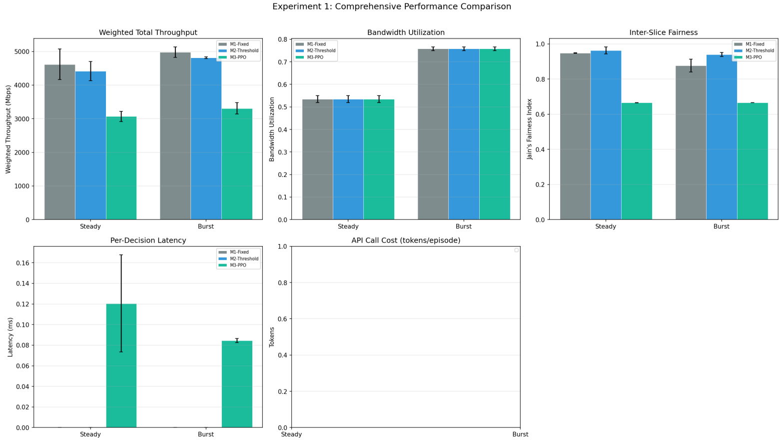


Burst event impact validation



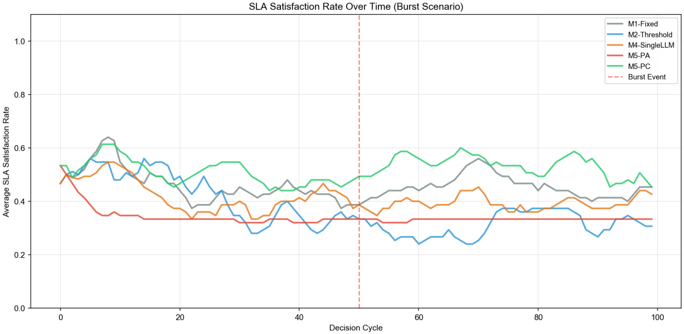
Inter-slice resource competition tradeoff

Appendix B: Experiment 1 — Comprehensive Performance Overview



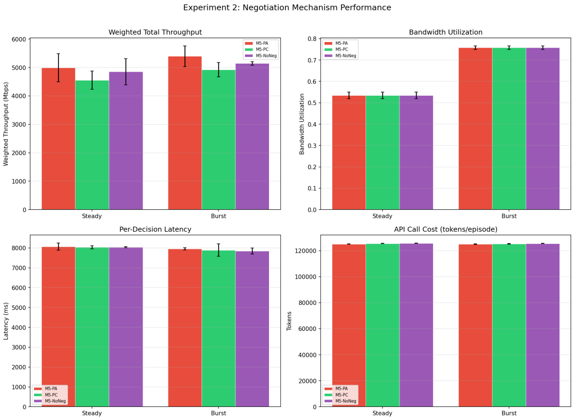
Throughput, bandwidth utilisation, fairness, decision latency, and API cost across all five methods

Appendix C: Experiment 1 — Burst Scenario SLA Trend



SLA satisfaction rate over decision cycles under burst scenario (eMBB users doubled at cycle 50)

Appendix D: Experiment 2 — Comprehensive Performance Overview



M5-PA / M5-PC / M5-NoNeg throughput, utilisation, latency, and API cost comparison